

CS 7646: Project 6 - Indicators and Technically Optimal Strategy

Andrew Myers
amyers72@gatech.edu

Abstract—Welcome to Joyner Document Format (JDF) v2.2! JDF is primarily intended to standardize page lengths while ensuring readability. Note that you are required to use JDF for all written assignments, but we will not perform explicit formatting checks. So, while improper formatting may be subject to penalties, you should not worry too much about whether your submission conforms to every minute detail; the most important elements are margins, font, font sizes, and line spacing. Just make a copy of one of the provided templates and replace its contents with your own, using the built-in paragraph styles.¹ If you do so, you do not need to verify that the style was followed.

1 INDICATORS

1.1 Simple Moving Average (SMA)

The Simple Moving Average indicator (SMA) tracks the mean price of an asset over N days. To calculate it over a period of time, we first retrieve the pricing data for the asset. When we do this, we opt to get data for extra days before the beginning of the specified period so that we can calculate the SMA for the first N days. We naively do this by subtracting $(N + 30)$ days from the start date, and use this as the first day of data to retrieve. Once we have all of the pricing data, we remove non-trading days (NaN) so they do not impact our indicators. Finally, we can calculate the rolling mean of prices over N days to get the SMA. To make the data easy to use, we can join the data against the date range to readd the NaN values. For these indicators, we are choosing to leave 'NaN' on non-trading days instead of forward filling. This is to ensure non-trading days do not impact our analysis.

SMA is a useful indicator to track the long-term trends of an asset. It is less sensitive to short-term fluctuations. An analyst could hypothesize that over time,

¹ Here are instructions for [Microsoft Word](#), [Apple Pages](#), and [Google Docs](#).

prices will revert to the mean. SMA would be a perfect indicator for this strategy. A simple implementation would be to buy when the price is below 95

1.2 Relative Strength Index (RSI)

The Relative Strength Index (RSI) is an indicator used to measure the speeds and changes of an asset's price movements. It does this by comparing the size of recent gains to recent losses. It uses a lookback period similar to SMA, so we start by retrieving price data earlier than the start date in the same way as described in 1.1. Once we have the pricing data, we calculate the pricing difference of each day from the previous day. We can then separate the "gains" from the "losses" based on whether that difference was positive or negative. Once we have the gains and losses, we calculate the averages for each of them over the lookback period N. Finally, we can calculate the RSI for each day using the formula: $RSI = 100 - (100 / (1 + (\text{average gain} / \text{average loss})))$. A value of 70 or above is generally considered overbought, and a value of 30 or below is generally considered oversold. Similar to SMA in 1.1, we are choosing to leave 'NaN' on non-trading days instead of forward filling.

RSI can be useful in trading strategies to help identify conditions where the asset is overbought or oversold. An analyst could use RSI along with SMA to try and identify situations where a new uptrend is beginning for an asset. To do this, they could buy when the RSI is below 30 (signalling oversold) and price crosses above the SMA. The opposite would be to sell when the RSI is above 70 (signalling overbought) and price crosses below the SMA.

1.3 Rate of Change (ROC)

The Rate of Change (ROC) indicator is used to measure the percentage change in price between the current price and the price N days ago. It can be used by analysts to measure how quickly the price of an asset is changing. To calculate the ROC, we first retrieve the pricing data for the asset in the same way as described in 1.1, with N + 30 extra days. Once we have the data, we can calculate the ROC for each day using the formula: $ROC = ((\text{price} - \text{price_N_periods_ago}) / \text{price_N_periods_ago}) * 100$. As we have done with the past indicators, we choose to remove 'NaN' during calculation and add them back in after the calculation is complete.

An analyst could choose to use ROC to signal strong trends in the asset's price. They may hypothesize that high ROC signals strong upward momentum, and low

ROC signals strong downward momentum. They could create a strategy that buys the asset when the ROC is above 10

1.4 Percentage Price Oscillator (PPO)

The Percentage Price Oscillator (PPO) is a momentum indicator that measures the difference between two moving averages. It is expressed as a percentage of the larger moving average. To calculate the PPO, we first need to choose two periods, one long and one short. We can then calculate the SMA of the asset's price over each of these periods using the SMA indicator from 1.1. Once we have the two SMAs, we can calculate the PPO using the formula: $PPO = 100 * (short_SMA - long_SMA) / long_SMA$.

PPO can be used to identify the momentum or trends of an asset's price. A positive PPO indicates that the short-term moving average is above the long-term moving average, which could be a bullish signal. Alternatively, a negative PPO could signal a downtrend and be a bearish signal. An analyst could use PPO to identify when the momentum of an asset is changing, and use this to inform their trading strategy. A simple strategy could be to buy when the PPO becomes positive and sell when the PPO becomes negative.

1.5 Fast Stochastic Oscillator (%K)

The Fast Stochastic Oscillator is a momentum indicator that compares the closing price of an asset to its price range over a specified period. It is expressed as a percentage called %K, and can be used to determine where a price is relative to its recent trading range. We calculate this indicator by first retrieving the pricing data for the asset, again using $N + 30$ extra days as described in 1.1 to ensure we avoid 'NaN' values caused by the lookback period. For each day, we then calculate the highest and lowest prices over the past N days. Finally, we can calculate the %K for each day using the formula: $\%K = 100 * ((price - lowest_price) / (highest_price - lowest_price))$. As with the other indicators, we choose to remove 'NaN' during calculation and add them back in after the calculation is complete.

An analyst could use this indicator as part of a strategy based on the hypothesis that during upward trends, the asset price will tend close relatively high. Alternatively, during downward trends, the asset price will tend to close relatively low. These trends could be determined using PPO described in 1.4. They could choose to buy the stock when %K is below 25% and PPO is positive with the idea that

over time %K will return to a higher value due to the upward trend. The opposite would be to sell when %K is above 75% and PPO is negative, signalling a downtrend.

2 THEORETICALLY OPTIMAL STRATEGY

In order to provide a benchmark to compare our indicators against, we implement an "optimal" strategy. This strategy does have the ability to look into the future before choosing to BUY or SELL. However, it is constrained in that it's only allowable positions are +1000, 0, and -1000 shares. We will initiate the benchmark with \$100,000, use "JPM" as the stock symbol, and use the time period January 1, 2008, to December 31, 2009. We also assume commission and impact are 0.

At a high level, the strategy simply buys as many shares as possible when the price will go up, and sells as many shares as possible when the price will go down. To implement the strategy, we iterate through each day and find that day's price as well as the next day's price. If the next day's price is higher than the current day's price, we move a position of 1000 shares (if not there already). If the next day's price is lower than the current day's price, we move a position of -1000 shares (if not there already). For each of these buy/sell cases, we update our cash total by subtracting the price of the shares bought/sold. In order to return the results in the form of an order list, each position movement is added to the list as a "BUY" or "SELL" order.

// TODO charts

3 DELETE THIS

All text in JDF should be set in the Palatino typeface. It is available practically everywhere as a system font: Google Docs offers a version called *Palatino Linotype*. It comes in regular and bold weights with matching italics.

Handgloves
Handgloves

Figure 1—Palatino. Make sure the live text (bottom) uses the same font as the image (top).

3.1 Body text

Body text is set in the regular weight at 11 points with custom line spacing of 1.26, and 8.5 points of spacing added after each paragraph. It should be justified and paragraphs should not be indented. These styles can be automatically applied using the *Normal* paragraph style.

Bold and *italics* should be used for emphasis. Hyperlinks may be inserted in the text, as well as in-line code, superscripts^{like this}, and subscripts_{like this}.

3.2 Title & subtitle

The paper title should be set in the regular weight at 17 points with 22 points of line spacing, centered at the top of the first page. The title may span up to three lines. For typical assignments, the document title may be as simple as "Assignment 1" More specialized assignments may warrant more unique paper names, like "A Proposal to Create a New Document Format."

The author's name and email should come next unless you want to or were asked to submit anonymously, in which case this can be omitted. They should be set in the same size and weight as body text, centered. These styles can be applied using the *Title* and *Subtitle* paragraph styles.

3.3 Abstract

If your paper requires an abstract, it should be placed at the top of the first page underneath the title block, preceded by the word *Abstract* in bold italic. An extra 0.5" should be added to both sides. Not all papers require abstracts; only those that would benefit from a high-level summary of the project or its background.

3.4 Headings

Headings should all be set in the same size as body text (11 pt) in bold. With the exception of *Heading 1*, they should all have 8.5 pt of space before and after. They should be hierarchically numbered: Word, Pages, and L^AT_EX will do this automatically when you use the appropriate paragraph styles, but Google Docs users will need to number their headings manually. Headings should not span more than one line.

3.4.1 *Heading 1*

Heading 1 should be set in all caps. It should have 11 points of space added before and 8.5 points of space added after

3.4.2 *Headings 2–4*

Besides *Heading 1*, which is set in caps, headings should always use sentence case (i.e., first word capitalized) rather than title case; after all, they are not titles. *Heading 2* should be set in bold roman (upright), and *Heading 3* should be set in bold italics. The use of headings beyond *Heading 3* is discouraged.

Heading 4—*Heading 4* is provided as a run-in sidehead. Like *Heading 3*, it is set in bold italics, but it should be followed by an em dash and flow right into the text, as seen at the beginning of the current paragraph. It should be used more as a list style than a heading style, e.g. to set off a list of principles in a heuristic evaluation.

3.5 Page layout

JDF uses the US Letter paper size (8.5" x 11"). It has a top margin of 1", and bottom and side margins of 1.5". This yields a text block of 5.5" x 8.5", which is exactly $\frac{1}{2}$ the size of the page, divided lengthwise.

The page number should be included in the bottom margin, 1" from the bottom of the page. No other elements should be placed in the margins.

4 PRESENTATIONAL ELEMENTS

You are encouraged to use presentational elements liberally, as long as they add to the clarity of your submissions. They often require less space and fewer accompanying words to explain a given concept, and do a far better job of it.

4.1 Figures

Figures should always be centered on the page, although they may also take up the entire width and height of the text block. Figures should always be referenced in the text, and they should include a descriptive caption. Figures may also be equations, diagrams, or other kinds of content.

If your figure includes a white background (e.g. an interface design or graph), it may aid legibility to add a $\frac{1}{4}$ -point black border.

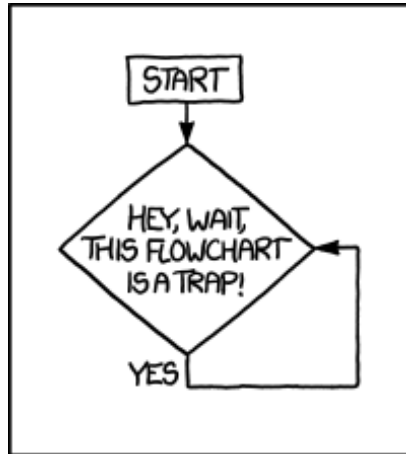


Figure 2—Make sure your flowcharts are more useful than this one. Source: [XKCD](#).

Figure captions should be placed beneath the corresponding figure, indented 1" on the left and right sides. The label for the figure, e.g. "Figure 1," should be set in bold italics followed by an em dash, and the entire caption should be 8.5 points with 14 points of line spacing. Again, Word and LaTeX will number these automatically using the *Figure Caption* paragraph style, but Docs users will need to number these manually. If need be, you may have one figure caption corresponding to multiple consecutive figures and use either locational descriptors (e.g. "top left," "middle") or labels (e.g. "A", "B") to map parts of the caption to parts of the figure. Make sure that caption falls on the same page as the corresponding figure or table; you may rearrange text to make this work.

In Microsoft Word, you may need to either change the image's text wrap settings to "Top and Bottom" or change the line spacing of the image to 1.0.

4.2 Tables

You have freedom to format tables in the way that works best for your data. Generally, text should be left-aligned and numbers should be right-aligned or aligned at the decimal – you can do this using a [custom tab stop](#). The default table style (shown below) reduces the text size to be equal to the caption text.

Table captions should be formatted the same way as figure captions, but they should be placed above the table. The popular mnemonic for this is: figures at the foot, tables at the top. Like figures, tables should not exceed the margins and should be centered on the page.

Table 1—Mathematical constants. Notice how the approximations align at the decimal.

Name	Symbol	Approximation	Description
Golden ratio	ϕ	1.618	Number such that the ratio of " to the number is equal to the ratio of its reciprocal to 1
Euler's number	e	2.71828	Exponential growth constant
Archimedes' constant	π	3.14	The ratio between circumference and diameter of a circle
One hundred	A+	100.00	The grade we hope you'll all earn in this class

4.3 Additional elements

There are additional elements you may want to include in your paper, such as in-line or block quotes, lists, and more. For other content types not covered here, you have flexibility in determining how it should be used in this format.

4.3.1 Quotes

If you would like to quote an outside source, you may do so with quotation marks followed by a citation. If a quote is fewer than three lines, you may write it in-line. It is acceptable to replace pronouns with their target in brackets for clarity. For example, "Heavy use of peer grading would compromise [the school's] reputation" (Joyner et al., 2016). If a quote exceeds three lines, you should set it as its own paragraph with 0.5" side margins, using the *Blockquote* style.

"Whether or not the grades generated by peers are reliably similar to grades generated by experts is only one factor worth considering, however. Student perception is also an important factor. A recent study indicated that reliance on peer grading is one of the top drivers of high MOOC dropout rates. This problem may be addressed by reintroducing some expert grading where possible." (Joyner et al., 2016)

4.3.2 Lists

Bulleted and numbered lists are indented 0.5" from the left margin, with the bullet or number hanging in the margin by 0.25" (the default format).

Bullet points:

- First bullet point item

- Second bullet point item

Numbered list:

1. First numbered item
2. Second numbered item

5 PROCEDURAL ELEMENTS

5.1 In-line citations

Articles or sources to which you refer should be cited in-line with the authors' names and the year of publication.² The citation should be placed close in the text to the actual claim, not merely at the end of the paragraph. For example: students in the OMSCS program are older and more likely to be employed than students in the on-campus program (Joyner, 2017). In the event of multiple authors, list them. For example: research finds sentiment analysis of the text of OMSCS reviews corresponds to student-assigned ratings of the course (Newman and Joyner, 2018). You may also cite multiple studies together. For example: several studies have found students in the online version of an undergraduate CS1 class performed equally with students in a traditional version (Joyner, 2018a; Joyner, 2018b). If you would like to refer to an author in text, you may also do so by including the year (in parentheses) after the author's name in the text. If a publication has more than 4 authors, you may list the first author followed by 'et al.' For example: Joyner et al. (2016) claim that a round of peer review prior to grading may improve graders' efficiency and the quality of feedback given. This applies to parenthetical citations as well, e.g. (Joyner et al., 2016).

5.2 Reference lists

References should be placed at the end of the paper in a dedicated section. Reference lists should be numbered and organized alphabetically by first author's last name. If multiple papers have the same author(s) and year, you may append a letter to the end of the year to allow differentiated in-line text (e.g. Joyner, 2018a and Joyner, 2018b in the section above). If multiple papers have the same author(s), list them in chronological order starting with the older paper. Only works that are

² In-line citations are preferred over footnotes, and we favor APA citation format for both in-line citations and reference lists. Refer to the [Purdue Online Writing Lab](#), or follow the above examples. Footnotes should use 8.5 point text with 1.26 line spacing.

cited in-line should be included in the reference list. The reference list does not count against the length requirements.

6 REFERENCES

1. Joyner, D. A., Ashby, W., Irish, L., Lam, Y., Langston, J., Lupiani, I., Lustig, M., Pettoruto, P., Sheahen, D., Smiley, A., Bruckman, A., and Goel, A. (2016). "Graders as Meta-Reviewers: Simultaneously Scaling and Improving Expert Evaluation for Large Online Classrooms". In: *Proceedings of the Third Annual ACM Conference on Learning at Scale*. Edinburgh, Scotland.
2. Joyner, D. A. (2017). "Scaling Expert Feedback: Two Case Studies". In: *Proceedings of the Fourth Annual ACM Conference on Learning at Scale*. Cambridge, Massachusetts.
3. Joyner, D. A. (2018a). "Intelligent Evaluation and Feedback in Support of a Credit-Bearing MOOC". In: *Proceedings of the 19th International Conference on Artificial Intelligence in Education*. London, United Kingdom: Springer.
4. Joyner, D. A. (2018b). "Toward CS1 at Scale: Building and Testing a MOOC-for-Credit Candidate". In: *Proceedings of the Fifth Annual ACM Conference on Learning at Scale*. London, United Kingdom: ACM Press.
5. Newman, H. and Joyner, D. A. (2018). "Sentiment Analysis of Student Evaluations of Teaching". In: *Proceedings of the 19th International Conference on Artificial Intelligence in Education*. London, United Kingdom: Springer.

7 APPENDICES

You may optionally move certain information to appendices at the end of your paper, after the reference list. If you have multiple appendices, you should create a section with a *Heading 1* of "Appendices." Each appendix should begin with a descriptive *Heading 2*; appendices can thus be referenced in the body text using their heading number and description, e.g. "Appendix 5.1: Survey responses." If you have only one appendix, you can label it with the word "Appendix" followed by a descriptive title, e.g., "Appendix: Survey responses."

These appendices do not count against the page limit, but they should not contain any information required to answer the question in full. The body text should be sufficient to answer the question, and the appendices should be included only for you to reference or to give additional context. If you decide to move content to an

appendix, be sure to summarize the content and note it in relevant place in the body text, e.g., "The raw data can be viewed in *Appendix 5.1: Survey responses*."